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Non-invasive, non-contact conductivity measurement using radiofrequency probe loading

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Abstract

Accurate measurement of electrical conductivity in water-based samples is essential for applications ranging from brine characterization in petroleum systems to biological sensing. Direct-contact methods are prone to electrode degradation, while indirect-contact methods often suffer from undefined current paths. This study introduces a non-invasive, non-contact technique for conductivity measurement in water-based samples using radiofrequency (RF) probe loading, leveraging inductive losses induced by eddy currents in conductive media. Changes in the quality factor of the RF probe are analyzed to establish a direct proportionality between inductive losses and sample conductivity. This relationship is validated across solenoidal, loop-gap resonator, and surface coil designs with cylindrical and planar homogeneous samples. Theoretical equations, derived from classical electromagnetism and the principle of reciprocity, closely align with experimental results, revealing the critical role of probe geometry and resonant frequency in measurement sensitivity. The findings demonstrate robust performance across diverse ionic solutions including samples with flow, achieving cost-effective measurement with minimal hardware. This method will allow differentiation between samples with differing conductivity, even when absolute conductivity values cannot be determined. This makes it especially suited for analyzing complex, heterogeneous materials like foods or biological samples, where localized average conductivity measurement in specific regions of the sample, may be used as a classification tool. The RF probe loading approach offers a versatile alternative to conventional conductivity measurement methods, with potential for real-time, non-invasive monitoring in dynamic systems.

Supplementary material for this article is available [online](#)

Keywords: RF probe loading, inductive loss, conductivity, surface coil, solenoidal probe, loop-gap resonator

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1. Introduction

Electrical conductivity measurement of water-based samples provides valuable information on ion concentration and mobility, making it essential in a wide range of applications, such as characterization of brines in petroleum systems [1] and vanadium redox flow batteries (VRFBs) for renewable energy storage [2]. In crude oil exploration [1], conductivity measurements help determine the formation water salinity in porous rock, an important factor in evaluating reservoir potential and guiding drilling decisions. In VRFB [2], electrolyte conductivity data is essential to improve battery efficiency, power output, and long-term maintenance strategies. In VRFBs, a non-contact conductivity measurement method is essential due to the corrosive nature of the electrolyte solution.

Existing conductivity measurements of water-based samples rely either on direct contact with the sample [3, 4] or on indirect coupling methods [5, 6]. Direct-contact sensors, such as electrochemical probes, require reference electrodes, are prone to fouling, and may require frequent recalibration [4, 6, 7]. In addition, undefined current paths can lead to variability in the cell constant, reducing measurement accuracy [8]. Indirect-contact methods based on inductive or capacitive coupling can mitigate some of these challenges [9–11]. However, inductive sensors often suffer from ambiguous current flow paths within the medium [12], while capacitive sensors tend to involve complex circuitry.

We present a simple, non-invasive, non-contact conductivity measurement technique based on radiofrequency (RF) probe loading, which relies on inductive interactions between a resonant circuit and the sample. The primary motivation of this study is to develop quantitative methods for measuring the conductivity of brine solutions with conductivity similar to that of water in oil reservoirs and salt water. Such measurements, are particularly relevant for applications where electrical conductivity is a useful bulk property; for instance, in the monitoring of salinity in porous media [1].

In magnetic resonance imaging (MRI), RF probe loading presents a challenge. Electromagnetically, the human body is a large volume of salty water, and when exposed to an alternating magnetic field, eddy currents heat the tissue. The fact that people vary in size and shape, further complicates RF probe design and performance for MRI. Hoult and Lauterbur [13] demonstrated that noise in magnetic resonance (MR) experiments with conductive samples depends on dielectric and inductive losses within the sample. Proper shielding and grounding can minimize dielectric losses [13–15], but no effective method exists to mitigate inductive losses. This study does not involve MR. Instead, it exploits RF probe loading with a simplified experimental setup to quantify inductive losses as a proxy for conductivity measurement.

The conductivity measurement concept is simple: an RF signal, essentially an alternating current (AC) oscillating at radio frequencies, flows through the RF probe. The RF probe is a tunable LC circuit designed to resonate at a specific frequency. The oscillating current in the RF probe generates an oscillatory magnetic field in the sample space. When a conductive sample is measured, this magnetic field induces eddy

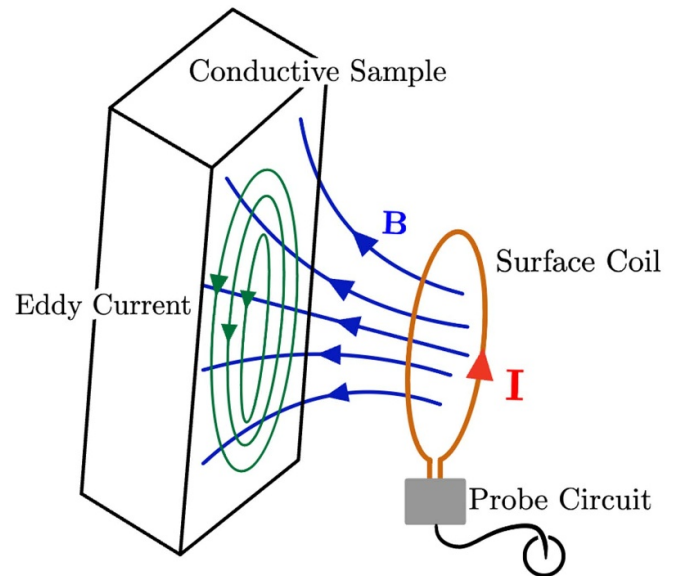


Figure 1. A simple schematic representation of a surface coil RF probe interacting with a conductive sample. An alternating current (I , shown in red) in the coil generates a magnetic field (B , shown in blue), which induces eddy currents (green loop) in the conductive sample. These eddy currents dissipate energy through resistive heating in the sample, increasing the RF circuit's effective resistance and altering the Q -factor of the circuit.

currents within the sample, which in turn dissipate power. Often referred to as inductive losses or eddy current losses, R_m , this energy dissipation reflects the sample's ability to absorb and dissipate electromagnetic energy. A simple schematic illustrating this concept is presented in figure 1, where a surface coil serves as the probe interacting with an adjacent planar conductive sample.

This study investigates how the quality factor (Q -factor) of an RF circuit responds to varying sample conductivities. The change in Q -factor upon introducing a sample in the RF probe represents an inductive loss, directly linking the losses and sample conductivity. To evaluate this approach, three distinct RF probe configurations were designed, characterized and tested: solenoidal coils and a loop-gap resonator (LGR) with cylindrical samples, and a surface coil with effectively infinite planar samples. The resonant behavior of each RF probe was analyzed using a network analyzer, enabling precise measurements of the Q -factor under different salinity conditions. Theoretical predictions of RF probe behavior were derived from first-principles electromagnetism and compared to experimental outcomes. The method's versatility was further explored by comparing flowing and stationary solution samples and evaluating its applicability to different salts in solution.

2. Theory and measurement of R_m

The Q -factor is commonly used to characterize a resonant circuit. This dimensionless parameter gauges the efficiency of a resonant circuit by comparing the energy it stores to the energy

it dissipates over one cycle of the RF signal [15, 16]. It is formally defined as

$$Q = \frac{\text{Maximum Energy Stored}}{\text{Average Energy Dissipated per Cycle}} = \frac{\omega_0 L}{R_{\text{eff}}}, \quad (1)$$

where $\omega_0 = 2\pi f_0$ is the resonant frequency, L is the inductance, and R_{eff} is the effective resistance, encompassing both intrinsic circuit losses and those induced by the sample. A high Q -factor signals a sharp resonance with minimal energy loss, while a lower Q -factor indicates a broader resonance due to increased dissipation as shown in figure 3(b). Changes in Q -factor directly reflect the electromagnetic interactions between the RF probe and the sample.

The effective resistance R_{eff} encapsulates all sources of energy loss in the RF probe and is expressed as

$$R_{\text{eff}} = R_{\text{coil}} + R_{\text{dielectric}} + R_{\text{radiative}} + R_{\text{m}}, \quad (2)$$

where each term reflects distinct physical mechanisms. The resistive loss, R_{coil} , arises from ohmic resistance in the probe's wires, amplified at high frequencies by the skin effect, which confines current to the conductor's surface [17]. To minimize this, low-resistance materials like copper are employed. Dielectric losses, $R_{\text{dielectric}}$, stem from electric fields fringing into the sample due to potential differences across the circuit, interacting with lossy dielectric materials [16]. In saline solutions, however, these are typically overshadowed by inductive losses, R_{m} , driven by eddy currents, as the sample's conductivity dominates [18]. Radiative losses, $R_{\text{radiative}}$, occur at higher frequencies [19] but remain minor in our setup compared to resistive and inductive contributions. Each term in equation (2) depends on the probe's design, its geometry and materials, and the operating frequency. In this work, the design prioritizes the sample's contribution, such that $R_{\text{eff}} \approx R_{\text{m}}$. This simplifies the Q -factor to $Q \approx \omega_0 L / R_{\text{m}}$, directly linking it to the inductive losses. These losses, driven by eddy currents, scale directly with the sample's electrical conductivity, σ , and we can express the relationships analytically. Detailed derivations of equations (3), (4) and (6) are provided in the supplementary materials.

2.1. Inductive losses for a solenoidal RF probe

Hoult and Lauterbur [13] quantified the inductive losses for a uniformly conducting spherical sample placed inside a solenoidal RF probe, as illustrated in figure 2(a). Employing the reciprocity theorem [20, 21] to evaluate the power dissipated in the sample due to a unit AC at the probe's resonant frequency, the inductive loss is given by

$$R_{\text{m}} = \frac{\pi \omega_0^2 \mu_0^2 n^2 b^5 \sigma}{30 (a^2 + g^2)}, \quad (3)$$

where a and g are the coil's radius and half-length, respectively, b is the sphere's radius, n is the number of coil turns, and μ_0 is the permeability of free space.

Hoult and Lauterbur [13] assumed a homogeneous magnetic field within the solenoid. This simplification is reasonable for tightly wound solenoidal RF probes where field variations are minimal near the coil center. Additionally, the skin effect is ignored, as the skin depth at radio frequencies is significantly larger than the characteristic dimensions of the sample across a wide range of conductivities. This ensures that induced currents span the entire sample volume.

Similarly, for a conducting cylindrical sample within a solenoidal RF probe (figure 2(b)), under the same simplifying assumptions, the inductive loss is

$$R_{\text{m}} = \frac{\pi \omega_0^2 \mu_0^2 n^2 b^4 g \sigma}{16 (a^2 + g^2)}, \quad (4)$$

where b is now the cylinder's radius. Other variables retain their prior definitions. Our derivation (see supplementary material) agrees with the work of Gadian and Robinson [22] and Cho [23], with Cho's formulation reducing to equation (4) despite using different variable conventions.

2.2. Inductive losses for a surface coil RF probe

The simplest RF coil designs are circular loops known as surface coils, which generate a magnetic field perpendicular to the coil plane, with field amplitude decreasing along the coil axis. In contrast to solenoids, where samples must fit within the coil, surface coils offer localized sensitivity that is well suited to planar or surface-constrained samples.

However, the strongly non-uniform magnetic fields produced by surface coils prevent the straightforward analytical approach applied in the solenoid case. Hayes and Axel [24] explored noise characteristics in flat surface coils. They conducted experiments with circular surface coils placed either in contact with, or near, a flat tank filled with a conductive saltwater solution, empirically measuring inductive losses as a function of coil radius. They proposed a functional form for sample resistance similar to that for solenoids,

$$R_{\text{m}} = \sigma \omega_0^2 f(a), \quad (5)$$

where σ represents the sample conductivity, ω_0 is the operating frequency, and $f(a)$ is an empirically determined function of coil radius a .

Extending the analysis from our earlier solenoidal RF probe study and addressing the challenges posed by the non-uniform field of surface coils, this work employed a magnetic vector potential approach presented in [25] to analytically derive the inductive loss caused by a planar conductive sample. The sample is assumed to be infinite laterally and in depth relative to the coil dimensions (see figure 2(c)). This simplifying assumption ensures maximal coupling between the coil's magnetic field and the conductive sample. The derivation involves surface integrals over the coil's area to rigorously account for the non-uniform field distribution. The resulting analytical expression for inductive loss is

$$R_{\text{m}} = \frac{1}{3} \omega_0^2 \mu_0^2 n^2 a^3 \sigma. \quad (6)$$

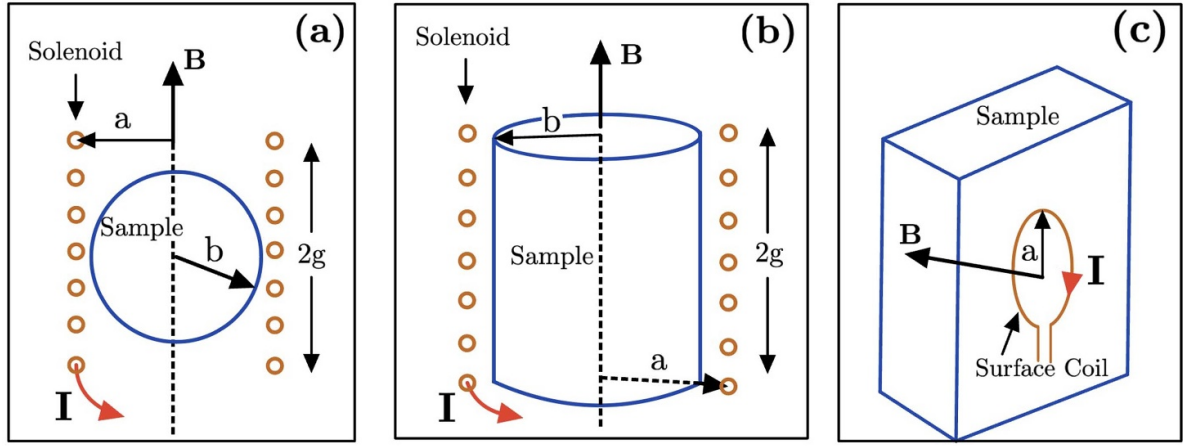


Figure 2. Schematic illustrations of conductive samples placed within RF probes, used for analytical calculations of inductive losses. (a) Spherical sample of radius b positioned centrally inside a solenoidal coil of radius a and length $2g$. (b) Cylindrical sample of radius b and length $2g$ placed centrally within the same solenoidal coil. (c) Planar conductive sample positioned adjacent to a surface coil of radius a .

where n is the number of turns of the coil and a is the radius of the surface coil. Note that, unlike equations (3) and (4), equation (6) is independent of sample size. This assumes that the samples are larger than the surface coil. Harpen [26] approached the problem similarly but used a summation approximation for an integral when calculating the magnetic vector potential A in spherical coordinates, employing an expansion in spherical harmonics. His approximation resulted in an inductive loss $R_m = 0.0332 \pi^2 \omega_0^2 \mu_0^2 n^2 a^3 \sigma$, which deviates from equation (6) by about 1%.

The analytical expressions derived for inductive losses scale directly with the conductivity, σ , the square of the resonant frequency, ω_0 , and strongly depend on characteristic size, the radius, b (for solenoids) or the coil radius, a (for surface coil). For a given probe design, examining equations (3), (4) and (6) indicates that, aside from the conductivity σ of the sample, all other parameters are fixed by probe and sample geometry and the chosen frequency. These equations therefore predict that the inductive loss should be directly proportional to the sample's conductivity, a relationship we seek to confirm experimentally.

The RF probes used in this work operate within a frequency range of 10–90 MHz. This frequency range was selected to balance sensitivity with manageable RF probe design complexity. Below 10 MHz, sensitivity of the RF probe deteriorates significantly because the efficiency of the resonant circuit scales with the square of the resonance frequency. Additionally, achieving resonance at lower frequencies requires larger inductors and capacitors, which introduce higher losses and reduce probe efficiency. Above 100 MHz, RF probe design becomes more challenging due to shorter wavelengths, which complicate circuit tuning and matching. Furthermore, reduced signal penetration from decreased skin depth, along with increased radiation losses, limits the effectiveness of RF probes for conductivity measurements at higher frequencies. The 10–90 MHz frequency range is a common frequency range for MR/MRI and a large body of literature exists describing suitable RF probes [14–16, 19, 27–29]. The

probe geometry determines its inductance, L , and since the resonant frequency of the LC circuit is given by $\omega_0 = 1/\sqrt{LC}$, the coil design was adjusted to achieve the required inductance, and an appropriate tuning capacitance, C , was selected to tune the probe to the desired frequency. At the same time, a suitable matching capacitor was chosen to ensure that the probe's impedance was matched to the system's characteristic impedance, typically a purely resistive 50Ω , in order to minimize signal reflection and maximize power transfer (see figure 3(a)).

Once the probe is tuned and matched, its Q -factor was determined. While several methods exist to determine the Q -factor, it is often evaluated using the reflection mode method (see figure 3(b)) as

$$Q = \frac{\omega_0 L}{R_{\text{eff}}} = \frac{2\omega_0}{\Delta\omega}, \quad (7)$$

when the probe is matched to the characteristic impedance (50Ω) [16]. Here, $\Delta\omega$ denotes the bandwidth at the -3 dB half-maximum reflected power level (at 0 dB, all the power is reflected, at -25 dB or lower all the power is considered transmitted).

By comparing the Q -factor with and without the sample, one can isolate the inductive loss term, R_m . Without the sample, the unloaded Q -factor, Q_u , is simply

$$Q_u = \frac{\omega_0 L}{R_{\text{extra}}} \Rightarrow \frac{1}{Q_u} = \frac{R_{\text{extra}}}{\omega_0 L}. \quad (8)$$

When the probe is loaded with a conductive sample, the Q -factor, Q_l , is

$$Q_l = \frac{\omega_0 L}{R_{\text{eff}}} = \frac{\omega_0 L}{R_m + R_{\text{extra}}} \Rightarrow \frac{1}{Q_l} = \frac{R_m + R_{\text{extra}}}{\omega_0 L}, \quad (9)$$

where $R_{\text{extra}} = R_{\text{coil}} + R_{\text{dielectric}} + R_{\text{radiative}}$. Subtracting equation (8) from equation (9), we isolate the inductive losses R_m as

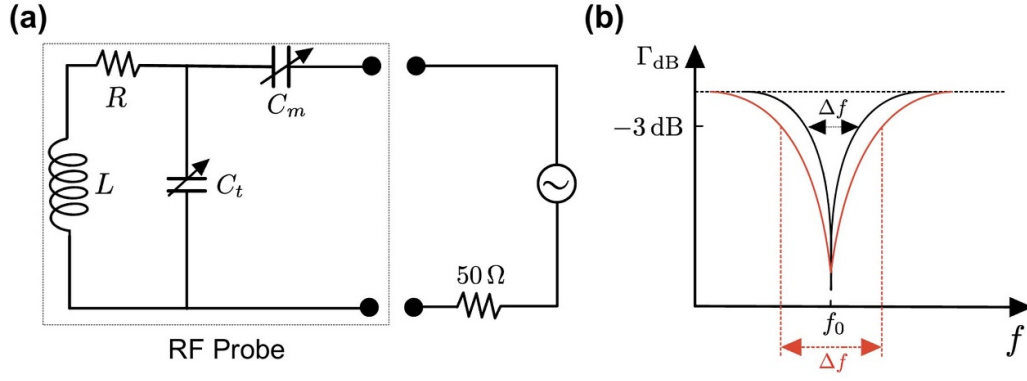


Figure 3. (a) Schematic of the parallel-tuned, series-matched LC circuit used for all probes in this work to achieve tuning and impedance matching. Here, L is the probe inductance, R is the resistance, C_t and C_m are the tuning and matching capacitors housed in the tuning box, and the circuit is matched to a $50\ \Omega$ source. (b) Measurement of the RF probe Q -factor using the reflection method. The $-3\ \text{dB}$ bandwidth of the response curve defines the Q -factor value. The sharp black curve represents an unloaded probe, while the broader red curve corresponds to a loaded condition due to conductive losses in the sample.

$$R_m = \omega_0 L \left(\Delta \frac{1}{Q} \right), \quad (10)$$

where $\Delta \frac{1}{Q} = \frac{1}{Q_l} - \frac{1}{Q_u}$. This expression shows that R_m is directly proportional to the change in $\Delta \frac{1}{Q}$ and also depends on the probe's inductance L and operating frequency ω_0 . Once inductive loss R_m is experimentally determined, it is related to the sample conductivity using equations (3), (4) and (6).

It is important to emphasize that in the frequency range 10–90 MHz, we operate in the near-field regime, which implies that radiative losses, $R_{\text{radiative}}$ are negligible compared to resistive and inductive contributions. Radiative losses are more important at higher frequencies. Also R_{coil} and $R_{\text{dielectric}}$ are considered negligible compared to the inductive losses, R_m .

3. Experimental

In most experiments, an RF network analyzer (300 kHz to 3000 MHz), model 8714B (Hewlett-Packard, Santa Clara, CA), was used for tuning, matching, and measuring the circuit's Q -factor. A portable handheld vector network analyzer (50 kHz to 1500 MHz), model NanoVNA-H (Seesii, Xincheng), was used for the same purpose in later experiments. Both the RF network analyzer and the portable NanoVNA have a frequency accuracy better than 1 kHz. An LC meter, model MLC500 (Dongguan Jingyan Instruments & Technology Co., Ltd. Kwai Chung, N.T.), was employed to measure the inductance of the RF probe circuits. The LC meter's specified accuracy is 1%. The inductance was determined from five repeated measurements, with the reported value representing their average. The accompanying uncertainty of the inductance is a combined uncertainty that accounts for both systematic contributions and statistical variation from repeated measurements [30], rather than a simple standard deviation.

The uncertainty calculation for inductive losses and conductivity was performed using standard propagation methods [30]. For inductive losses, the uncertainty was calculated by combining the uncertainties of the quality factors and inductance of the RF probe. For the conductivity, the uncertainty

was calculated combining uncertainties from both the slope and intercept of the fitted data in the inductive losses vs conductivity plot.

3.1. Solenoidal RF probe

A ten-turn solenoid, 50 mm in length and 50.3 mm in diameter, wound from a 16-gauge copper wire, was tuned and matched to 10 MHz using variable capacitors and a 5 pF fixed high- Q air-spaced capacitor on the tuning side. The RF circuit was housed in a separate tuning box, which served as a Faraday cage to prevent electric field lines. Eliminating the electric fields eliminates dielectric losses. However, its proximity to the RF coil may introduce a small additional loss. The experimental setup is shown in figure 4.

The coil's inductance, measured with an LC meter, was $3.42 \pm 0.04\ \mu\text{H}$. This value, taken without the sample, serves as our baseline for calculating inductive losses. Introducing a conductive sample slightly reduces the effective inductance due to eddy currents, but this reduction, typically 5%–10% in RF coils [31], falls within our measurement uncertainty and minimally impacts R_m . This approach aligns with standard RF resonator analysis where inductance variations are small compared to resistive effects [32].

A cylindrical sample container (outer diameter 40 mm, length 64.5 mm) was placed within the solenoid, and the circuit's Q -factor was measured for samples with varying sodium chloride concentrations, including distilled water. The dimensions of the container were chosen to match those of the RF probe in order to maximize the inductive response of the circuit. We note that equation (4) shows the measurement is very sensitive to sample size, with a dependence on the fourth power of the radius.

3.2. LGR

The LGR was constructed by wrapping a 0.025 mm thick, 50 mm $\times$ 157 mm copper foil around a cylindrical plastic form with an outer diameter of 50 mm, an inner diameter

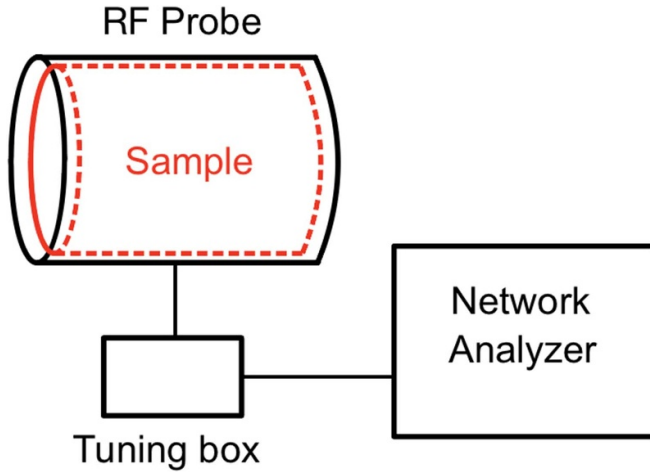


Figure 4. Schematic of the experimental setup for conductivity measurement via RF probe loading. A cylindrical sample is placed inside a probe (either a solenoid or a loop-gap resonator) connected to a tuning box and a benchtop network analyzer. Each probe has an outer diameter of ~ 50 mm, an inner diameter of ~ 43.6 mm, and a length of ~ 50 mm. The dimensions of the sample container were chosen to match those of the probe.

of 43.6 mm, and a length of 60 mm. The gap between the two ends of the foil was approximately 0.4 mm. The probe's inductance, measured across the two terminals connecting it to the tuning box using an LC meter, was $0.085 \pm 0.003 \mu\text{H}$.

Hornak *et al* [28] designed similar LGRs for MR using a single-turn copper loop mounted on a plastic sleeve, allowing for impedance matching by adjusting the loop's distance from the resonant cylinder. In their approach, tuning capacitors were placed directly across the gap in the copper cylinder. In this work, the RF probe was connected directly to a tuning box containing variable capacitors for both tuning and matching (figure 4). This parallel-tuned, series-matched configuration [16] was successful, with precise tuning to 90 MHz and stable impedance matching to 50Ω across all tested conditions.

3.3. Surface coil RF probe

A two-turn surface coil with a 1 cm radius and an inductance of $0.344 \pm 0.002 \mu\text{H}$ was constructed from a 24-gauge copper wire, tuned to 50 MHz, and matched to 50Ω using variable capacitors. Fixed capacitors of 15 pF and 20 pF were incorporated on the series matching and parallel tuning sides, respectively.

3.4. Solenoidal RF probe for flow

A solenoid with eight turns, 3.5 cm in length, and a radius of 2.24 cm was wound from a 16-gauge copper wire, yielding an inductance of $0.76 \pm 0.04 \mu\text{H}$. Integrated into an LC circuit tuned to 24 MHz, it included fixed capacitors of 15 pF (matching) and 20 pF (tuning) within a tuning box. The compact NanoVNA-H replaced the bulkier 8714B Network Analyzer, enabling portable and space-efficient measurements.

4. Results and discussions

4.1. Solenoidal RF probe with NaCl solution samples

This experiment investigated whether the measured inductive losses agrees with the theoretically predicted linear dependence on conductivity, as expressed in equation (4). Five NaCl solution samples with a salinity range of 1%–4%, including distilled water, were prepared in a cylindrical sample container (outer diameter 40 mm, length 64.5 mm). For context, freshwater typically contains less than 0.05% salt, while seawater averages 3.5% salinity [33, 34].

Table 1 demonstrates the influence of increasing NaCl concentration on RF probe loading. As salinity increases, the Q -factor decreases due to increased conductive losses, which directly correspond to an increase in inductive loss R_m . Table 2, on the other hand, compares the total effective resistance, $R_{\text{eff}} = \frac{\omega_0 L}{Q}$ with the inductive losses R_m , obtained from equation (10), for NaCl solutions fully inserted into the RF coil. The table also includes theoretical inductive losses, $R_{m,\text{theory}}$ calculated using equation (4), which assumes uniform sample placement. It is worth noting that solid NaCl did not produce any observable loading effect. Despite the presence of charge in solid NaCl, there is no mobility in the solid state; hence, there is neither conductivity nor inductive loss.

The conductivity at each salinity level was determined from the slope and intercept of a conductivity-versus-salinity calibration curve for NaCl solutions [35]. However, the high conductivity value (y-intercept) at 0% salinity, based on this method, requires scrutiny, as the data in [35] begins at 0.5% salinity. To address this, we adopted the distilled water conductivity value of $0.04193 \mu\text{S cm}^{-1}$ from [36]. All literature conductivity values are referenced at 20°C .

Figure 5(a) illustrates the relationship between inductive loss, R_m and conductivity, σ of NaCl solutions. The theoretical fit using equation (4) yields a slope of $0.221 \pm 0.003 \Omega\text{m S}^{-1}$, with an intercept of $0.02 \pm 0.01 \Omega$. The non-zero intercept results in a small overestimation of conductivity at 0% salinity, likely due to the sample container's contribution plus minor systematic offsets. The standard deviation of the residuals is 0.02Ω , comparable to the measurement uncertainty (i.e. 0.02 – 0.03Ω), indicating a reasonable fit despite potential asymmetries in sample placement. This agreement validates equation (4) for estimating inductive losses under similar conditions.

It is important to consider the sensitivity of the proposed method. We do so by considering the slope of plots like figures 5 and 7. Using the uncertainty of the slope and the intercept, we may calculate the uncertainty in conductivity and the sensitivity of the method. The method's absolute uncertainty, estimated from conductivity measurements of 1% (1.9 ± 0.1 mS) and 4% (6.2 ± 0.1 mS) brine solutions, is approximately 0.135 mS. This uncertainty corresponds to relative errors of 6% and 2.5% for the 1% and 4% brine solutions. The detection limit for achieving less than 10% relative error is approximately 1.35 mS, which precludes the reliable measurement of low-conductivity samples such as drinking water.

Table 1. Measured frequency bandwidth at -3 dB (Δf), Q -factors, inverse Q -factor differences ($\Delta(1/Q)$), and inductive losses (R_m) for NaCl solutions of varying salinity (in mass percent), using a solenoidal RF probe tuned and matched at 10 MHz. The measured inductance with the calculated uncertainty of the RF probe is $L = 3.42 \pm 0.04$ μ H.

NaCl density (%)	Δf (MHz) ± 0.001	Q -factor	$\Delta(1/Q)$ ± 0.0001	R_m (Ω)
Empty probe	0.098	204 ± 3	...	...
0	0.100	200 ± 3	0.0001	0.02 ± 0.02
1	0.140	143 ± 1	0.0021	0.45 ± 0.02
2	0.171	117 ± 1	0.0037	0.78 ± 0.02
3	0.197	102 ± 1	0.0050	1.06 ± 0.02
4	0.228	88 ± 1	0.0065	1.40 ± 0.03

Table 2. Comparison of the total effective resistance R_{eff} with measured inductive losses R_m , obtained from equation (10), for NaCl solutions fully inserted into the 10 MHz solenoidal RF probe. Experimental values are compared to theoretical predictions from equation (4).

NaCl density (%)	Experimental			$R_{m,\text{theory}}$ (Ω)
	R_{eff} (Ω)	R_{extra} (Ω)	R_m (Ω)	
0	1.13	1.11	0.02	0.00
1	1.58	1.13	0.45	0.45
2	1.93	1.15	0.78	0.76
3	2.23	1.16	1.06	1.08
4	2.58	1.18	1.40	1.40

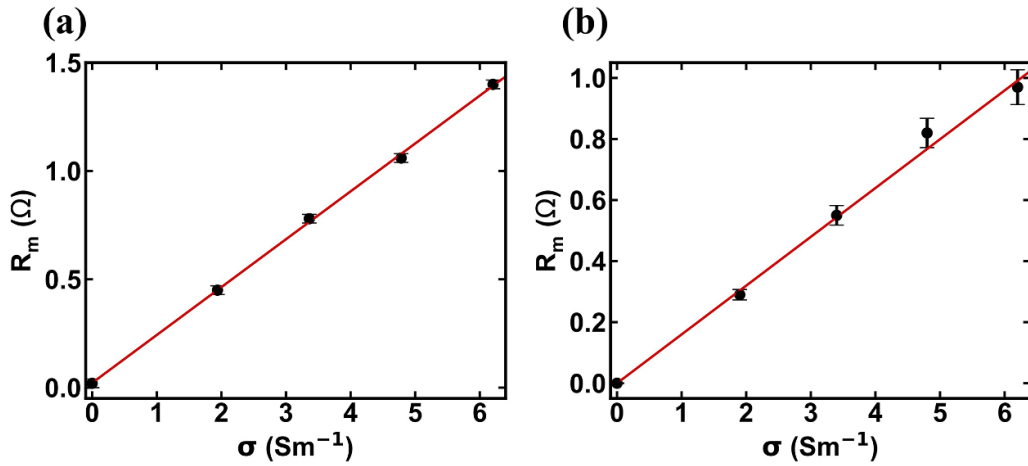


Figure 5. Summary of inductive loss R_m vs conductivity σ across various probe configurations. Literature conductivity values plotted on the horizontal axis are taken from [35, 36]. (a) Solenoidal RF probe tuned to 10 MHz with cylindrical NaCl solution samples, slope 0.221 ± 0.003 $\Omega\text{m S}^{-1}$, intercept 0.02 ± 0.01 Ω . (b) Loop-gap resonator RF probe tuned to 90 MHz with cylindrical NaCl solution samples, slope $m = 0.16 \pm 0.01$ $\Omega\text{m S}^{-1}$, intercept 0.00 ± 0.03 Ω .

4.2. LGR with NaCl solution samples

As established in equations (4) and (6), the inductive loss is proportional to the square of the resonant frequency ($R_m \propto \omega_0^2$). This relationship suggests that increasing the probe's frequency enhances sensitivity to conductivity variations. While solenoidal RF probes provide uniform magnetic fields and strong coupling, increasing their resonant frequency requires reducing either inductance or capacitance ($\omega_0 = 1/\sqrt{LC}$). However, reducing capacitance may limit tuning and matching, making inductance the primary parameter to adjust.

Achieving a significant reduction in L while maintaining sample compatibility and field homogeneity is challenging.

A promising alternative is the LGR [28, 37], which offers lower inductance while preserving B-field uniformity. LGRs function as single-turn solenoids with integrated capacitive gaps, naturally achieving high resonance frequencies without excessive reductions in capacitance. They have been widely adopted in electron spin resonance and high-frequency MR due to their high Q -factors and efficient sample coupling [29, 37]. To assess the sensitivity enhancement in inductive loss measurements, an LGR with the same dimensions as

Table 3. Measured frequency bandwidth at -3 dB (Δf), Q -factors, inverse Q -factor differences ($\Delta(1/Q)$), and inductive losses (R_m) for NaCl solutions of varying salinity (in mass percent), using a loop-gap resonator (LGR) RF probe tuned and matched at 90 MHz. The measured inductance with the calculated uncertainty of the RF probe is $L = 0.085 \pm 0.003 \mu\text{H}$.

NaCl density (%)	Δf (MHz) ± 0.001	Q -factor	$\Delta(1/Q)$ ± 0.00001	R_m (Ω)
Empty probe	2.081	86.5 ± 0.1	...	...
0	2.082	86.5 ± 0.1	0.00 001	0.000 ± 0.001
1	3.180	56.60 ± 0.03	0.00 610	0.30 ± 0.02
2	4.145	43.4 ± 0.01	0.01 147	0.55 ± 0.03
3	5.150	35.00 ± 0.01	0.01 705	0.82 ± 0.05
4	5.720	31.50 ± 0.01	0.02 020	0.97 ± 0.06

the 10 MHz solenoidal probe (section 4.1) was designed and tested.

The measured data in table 3 show a clear trend: as salinity increases, the Q -factor decreases while the inductive loss increases. Figure 5(b) plots the inductive losses for the LGR probe against NaCl solution conductivity, showing a positive linear relationship with a slope $m = 0.16 \pm 0.01 \Omega\text{mS}^{-1}$ and an intercept of $0.00 \pm 0.03 \Omega$, derived from a least-squares fit to $R_m = m\sigma + c$ (with c being the intercept). In contrast, figure 5(a) presents the inductive losses for the 10 MHz solenoidal probe, with a slope of $0.221 \pm 0.003 \Omega\text{mS}^{-1}$.

Contrary to expectations, the LGR's slope is slightly lower than that of the solenoid, despite a frequency jump from 10 MHz to 90 MHz. Note that theoretically, $R_m \propto \omega_0^2 \sigma \int_V B^2 dV$, where $\int_V B^2 dV$ represents the integral of the squared magnetic field over the sample volume ([13, 16]), with losses determined from $R_m = \omega_0 L \left(\Delta \frac{1}{Q} \right)$. The frequency ratio, $(90/10)^2 = 81$, suggests a substantial increase in slope. However, the LGR's single-turn design produces a weaker magnetic field ($\int_V B^2 dV$) than the multi-turn solenoid, as B scales with turn count [31]. Its lower inductance ($0.085 \mu\text{H}$ vs $3.420 \mu\text{H}$) further tempers R_m sensitivity to σ . This offsets the ω_0^2 gain, yielding the observed slope. While the LGR's 90 MHz operation ensures uniform field distribution, its reduced sensitivity reflects a trade-off between frequency and coupling strength, suggesting geometry optimization (e.g. increasing $\int_V B^2 dV$ through design adjustments) to better harness the frequency advantage [32].

The reduced sensitivity of the LGR is confirmed by a sensitivity analysis performed in the same manner as for the solenoid RF probe. From the conductivity measurements of 1% ($1.9 \text{ mS} \pm 0.2 \text{ mS}$) and 4% ($6.0 \text{ mS} \pm 0.4 \text{ mS}$) brine solutions, the absolute uncertainty is estimated to be 0.3 mS . This uncertainty corresponds to relative errors of 10% and 6% for the 1% and 4% brine solutions, respectively; both of which are higher than those observed with the solenoid RF probe. As with the solenoid, accuracy improves with increasing conductivity; however, for the LGR, the detection limit for achieving a relative error of less than 10% is approximately 2.85 mS . Compared to the solenoid, the LGR exhibits roughly double the relative errors, indicating greater measurement variability and a higher detection threshold.

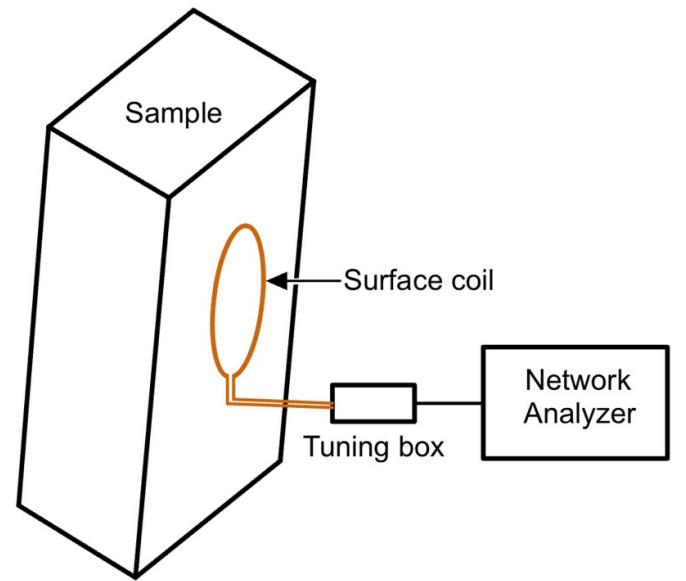


Figure 6. Schematic of the experimental setup showing a planar sample (600 ml cell culture flask, approximately $10.5 \text{ cm} \times 3 \text{ cm} \times 14 \text{ cm}$) positioned adjacent to a surface coil RF probe (radius 1 cm). The coil is tuned to 50 MHz and connected to a tuning box, which is in turn connected to a benchtop 8714B network analyzer for Q -factor measurement.

4.3. Surface coil RF probe with NaCl solution samples

This experiment evaluated whether the observed inductive losses agrees with the theoretically predicted linear dependence on conductivity, as described by equation (6). Five NaCl solution samples with a salinity range of 1%–4%, including distilled water, were prepared in a 600 ml cell culture flask with approximate dimensions of $10.5 \times 3 \times 14 \text{ cm}$. The lateral dimensions of the flask are much larger than the RF coil radius and can therefore be regarded as effectively infinite relative to the coil size.

The RF coil was mounted in contact with the center of one vertical side of the flask, oriented such that the normal to the coil's plane was perpendicular to the flask's surface, aligning the time-varying magnetic field perpendicular to the surface to maximize field penetration into the sample (see figure 6). The Q -factor of the RF circuit was measured for each sample.

Table 4 presents the measured frequency shifts (Δf), Q -factors, inverse Q -factor differences ($\Delta(1/Q)$), and inductive

Table 4. Measured frequency bandwidth at -3 dB (Δf), Q -factors, inverse Q -factor differences ($\Delta(1/Q)$), and inductive losses (R_m) for NaCl solutions of varying salinity (in mass percent), using a surface coil RF probe tuned and matched at 50 MHz. The measured inductance with the calculated uncertainty of the RF probe is $L = 0.344 \pm 0.002 \mu\text{H}$.

NaCl density (%)	Δf (MHz) ± 0.001	Q -factor	$\Delta(1/Q)$ ± 0.00002	R_m (Ω)
Empty probe	0.676	148.0 ± 0.3	...	...
0	0.738	135.5 ± 0.3	0.00 060	0.067 ± 0.004
1	1.250	80.2 ± 0.1	0.00 570	0.62 ± 0.04
2	1.686	59.30 ± 0.05	0.01 010	1.1 ± 0.1
3	2.104	47.50 ± 0.03	0.01 430	1.5 ± 0.1
4	2.453	40.80 ± 0.02	0.01 780	1.9 ± 0.1

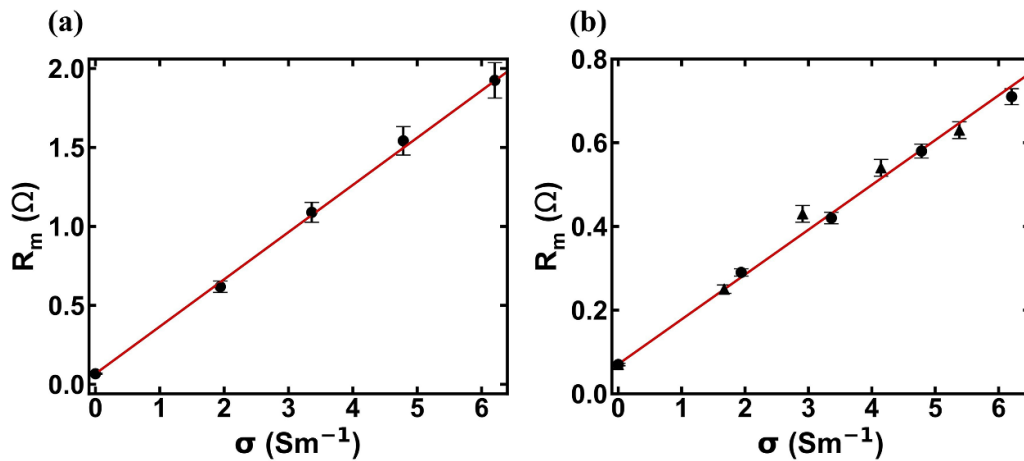


Figure 7. Summary of inductive loss R_m vs conductivity σ across various probe configurations. Literature conductivity values plotted on the horizontal axis are taken from [35, 36]. (a) Surface coil RF probe tuned to 50 MHz with planar NaCl solution samples, slope $0.30 \pm 0.01 \Omega \text{ m S}^{-1}$, intercept $0.070 \pm 0.004 \Omega$. (b) Surface coil RF probe tuned to 50 MHz comparing NaCl and CaCl_2 solutions, slope $0.110 \pm 0.002 \Omega \text{ m S}^{-1}$, intercept $0.100 \pm 0.002 \Omega$.

losses (R_m) for NaCl solutions of varying salinity, consistent with the approach outlined in section 4.1. Figure 7(a) shows a strong correlation between inductive losses, R_m and conductivity, σ for NaCl solutions up to 4% salinity. This is reinforced by the excellent fit of equation (6) to experimental data, confirming $R_m \propto \sigma$ with a slope of $0.30 \pm 0.01 \Omega \text{ m S}^{-1}$ and an intercept of $0.070 \pm 0.004 \Omega$.

Compared to the 10 MHz solenoidal probe (slope $0.221 \pm 0.003 \Omega \text{ m S}^{-1}$, section 4.1), the surface coil at 50 MHz shows a 25% increase in slope, consistent with the expected frequency scaling ($(50/10)^2 = 25$), though moderated by differences in $\int_V B^2 dV$ due to the surface coil's less uniform field distribution. The standard deviation of the residuals between the measured data and the fitted curve is 0.04Ω , indicating reasonable model accuracy despite a slight non-zero intercept (approximately 0.07Ω). This intercept, observed at 0% salinity, is likely due to the minor conductivity of distilled water ($\sim 10^{-6} \text{ S m}^{-1}$) from impurities or dissolved CO_2 , which is not surprising given typical water purity levels. Otherwise, this agreement validates equation (6) for surface coil configurations, supporting its applicability in RF probe loading studies.

Another important distinction between the surface coil RF probe and the solenoidal RF probe lies in the dependence of their inductive losses on sample size. For the solenoidal RF probe, the inductive loss exhibits a pronounced dependence on

sample dimensions, scaling with the fourth power of the cylindrical sample's radius (as shown in equation (4)). By contrast, the inductive loss of the surface coil RF probe is independent of sample size (see equation (6)), assuming a sample that is large compared to the RF probe. Sample size independence is a key advantage of the surface coil RF probe, as it enables conductivity measurements on large samples while minimizing the impact of sample shape and inhomogeneity. Moreover, the measurement remains spatially localized, which is advantageous for sample classification with different sizes, such as distinguishing between soft- and hard-shell lobsters, which is to say lobster maturity [38]. In this case, although the RF probe covers the same surface area, the effective inductive loss (saline water content sensed by the RF probe), we hypothesize, will differ depending on lobster maturity, providing a reliable classification parameter.

4.4. Surface coil RF probe with CaCl_2 solution samples

In experiments described earlier, we showed the efficacy of a non-invasive, non-contact RF probe loading method for measuring the conductivity. To extend this approach, we wish to confirm its applicability to other salt solutions. In this experiment, we conducted conductivity measurements for calcium chloride (CaCl_2) solutions and compared the results with

Table 5. Comparison of conductivity values from literature [35, 36], DDS-307 electrode-based desktop conductivity meter measurements, and our RF probe loading technique for NaCl and CaCl₂ solutions at various concentrations in mass percent.

Density (%)	Literature (Sm ⁻¹)		Desktop meter (Sm ⁻¹)		Probe loading (Sm ⁻¹)	
	NaCl	CaCl ₂	NaCl	CaCl ₂	NaCl	CaCl ₂
1	2.0	1.7	1.7	1.7	2.1 ± 0.1	1.7 ± 0.1
2	3.4	3.0	3.2	3.2	3.3 ± 0.2	3.4 ± 0.2
3	4.8	4.1	4.4	4.6	4.8 ± 0.3	4.4 ± 0.3
4	6.2	5.4	6.0	6.0	6.0 ± 0.3	5.2 ± 0.3

those of NaCl solutions. Since the RF signal losses R_m in the receiving coil are directly linked to the sample's conductivity σ ($R_m \propto \sigma$), rather than the specific type of ions present, we anticipated that the results for CaCl₂ solutions will agree with those for NaCl solutions.

The same surface coil RF probe from section 3.3, tuned to 50 MHz, was used in this experiment. Five samples each of NaCl and CaCl₂ solutions, including distilled water, were prepared in 450 ml cell culture flasks. Sample conductivity was also measured with a DDS-307 desktop conductivity meter (Hefei Chenchuang Instrument Co., Ltd. Hefei), an electrode-based device. For each sample, the Q -factor of the RF probe was measured in both unloaded and loaded conditions.

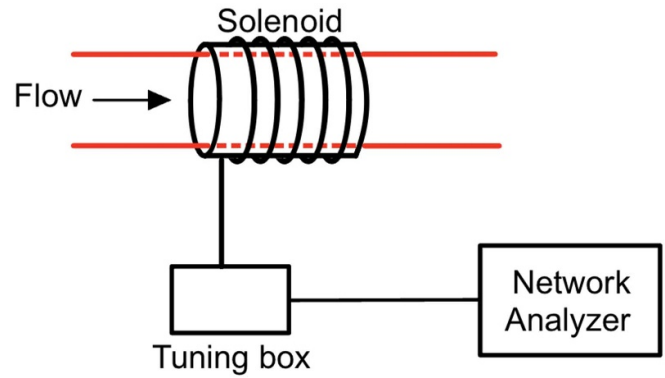
Table 5 compares the conductivity values from the literature [35, 36], measurements with the desktop EC meter, and the RF probe loading technique. The RF probe loading conductivity values were calculated by rearranging equation (6):

$$\sigma = \frac{3R_m}{\omega_0^2 \mu_0^2 n^2 a^3}, \quad (11)$$

where μ_0 is the permeability of free space, n is the number of turns of the coil, a is the radius of the surface coil, and σ is the conductivity of the sample. The close agreement between the conductivity values measured using the desktop EC meter, literature values, and those obtained from the RF probe loading method is encouraging.

Figure 7(b) presents the relationship between inductive losses, R_m and conductivity, σ for NaCl and CaCl₂ solutions. A least-squares fit to the combined data, using equation (6), yields a slope of $0.110 \pm 0.002 \Omega \text{m S}^{-1}$, lower than the $0.30 \pm 0.01 \Omega \text{m S}^{-1}$ from NaCl alone (section 4.3). This decrease mainly arises from differences in sample containers enclosed with the surface coil [31]. The earlier experiment used a low-density polyethylene (LDPE) container with a wall thickness of $0.110 \pm 0.002 \text{ mm}$, while the later used a polystyrene container with a thicker wall, $1.50 \pm 0.02 \text{ mm}$, causing the coil's non-uniform magnetic field to decay further before coupling with the sample. Although the container face touches the coil, the thicker wall increases the sample's distance from the field's peak, reducing R_m per unit conductivity.

The standard deviation of the residuals between the measured data and the fitted curve is 0.02Ω , indicating reasonable model accuracy though a slight non-zero intercept ($0.070 \pm 0.002 \Omega$) is observed here as well. The close alignment of R_m for both salts confirms that inductive losses depends on conductivity, not ion type, supporting the RF probe loading method's applicability across different ionic solutions.

**Figure 8.** Schematic of the experimental setup for conductivity measurement of flowing samples. A sample flows through a plastic pipe (inner radius $\sim 2 \text{ cm}$) positioned within a solenoidal coil (radius 2.24 cm , length 3.5 cm). The coil is part of an LC circuit housed in a tuning box and connected to a portable, handheld NanoVNA-H for Q -factor measurement.

4.5. Conductivity measurement of flowing samples

To evaluate the suitability of the RF probe loading technique for flow applications, we measured the conductivity of flowing solutions. The electrical conductivity of an ionic fluid, such as a salt solution, is governed by the concentration and mobility of its charge-carrying ions, properties that remain invariant under uniform flow relative to a stationary state, provided temperature and composition are constant [39]. This invariance holds because conductivity arises from microscopic ionic motion, which is unaffected by macroscopic fluid dynamics under typical laminar conditions [40]. However, measuring conductivity in flowing samples poses challenges due to potential artifacts such as mixing, electrode polarization, or flow-induced noise that may distort traditional contact-based methods. This motivates testing whether our non-invasive RF probe loading technique maintains accuracy in dynamic conditions comparable to static ones. If validated across varying flow rates, this method could enable real-time conductivity monitoring in fluid systems, surpassing the limitations of conventional probes. Here, we assess the solenoidal RF probe's performance in measuring the conductivity of a salt solution at controlled flow rates with minimal turbulence, using static measurements as a reference.

The fluid under test flowed through a Fisherbrand pipe (inner radius $\approx 2 \text{ cm}$) centered in the solenoid (see figure 8), driven by a gravity-fed system with a valve to regulate flow

Table 6. Inductive losses (R_m) and Q -factors for tap water and 2% saline solution at varying flow rates, compared to the unloaded probe. The uncertainty in Δf is 0.001 MHz.

Flow rate (LPM)	Tap water			2% Saline		
	Δf (MHz)	Q	R_m ($\pm 0.03 \Omega$)	Δf (MHz)	Q	R_m ($\pm 0.1 \Omega$)
Unloaded	0.30	160 ± 1	...	0.30	160 ± 1	...
0.0	0.42	114.3 ± 0.4	0.60	0.69	69.6 ± 0.1	2.0
0.5	0.42	114.3 ± 0.4	0.60	0.69	69.6 ± 0.1	2.0
1.8	0.42	114.3 ± 0.4	0.60	0.69	69.6 ± 0.1	2.0
3.0	0.42	114.3 ± 0.4	0.60	0.69	69.6 ± 0.1	2.0

rates from 0 to 3 liters per minute (LPM). We tested two samples: tap water (baseline) and a 2% NaCl solution (saline test case).

The experimental procedure was straightforward. Flow rates were determined by timing the filling of a 500 ml cylinder. To ensure laminar flow conditions, flow rates were maintained between 0 and 3 LPM through a 4 cm diameter pipe, yielding Reynolds numbers from 0 to 1592, which are well below the turbulent threshold of 2000 for both tap water and the 2% saline solution at approximately 20 °C. The unloaded solenoid's Q -factor was recorded prior to sample introduction. For each sample (tap water and 2% saline) the Q -factor was measured at each flow rate.

Table 6 presents the inductive losses, R_m for tap water and a 2% saline solution at varying flow rates. The flow-rate independence was anticipated but experimental proof was reassuring. Compared to contact-based meters, which struggle with flowing samples, this non-invasive approach offers a practical advantage for real-time applications.

5. Conclusion

This study successfully demonstrated the viability of a non-invasive, non-contact method for determining the electrical conductivity of water-based samples using RF probe loading, which relies on inductive losses caused by eddy currents. By analyzing the quality factor response of RF probes to varying sample conductivities, we established a direct correlation between inductive loss, R_m , and conductivity, σ , as predicted by analytical expressions. Experiments with solenoidal probes, a LGR, and a surface coil across both cylindrical and planar homogeneous samples validated this proportionality, aligning closely with theoretical expectations despite minor deviations due to magnetic field distribution and sample volume effects. The consistency of R_m values between NaCl and CaCl₂ solutions confirmed that inductive losses depend on conductivity not ion type, confirming the method's generality. Preliminary tests with flowing samples revealed no dependence on flow rate under laminar conditions, suggesting potential for real-time monitoring applications. Furthermore, the use of a NanoVNA made the setup highly cost-effective, with a total system cost of approximately \$300 CAD. Our work demonstrates that one can analytically determine the conductivity of solutions under test with RF probe loading.

While our current conductivity measurement method is based on uniform liquid samples to establish a quantitative

baseline, the technique is designed with the long-term goal of also being applied to inhomogeneous or irregularly shaped samples, where precise spatial mapping is not required, and average conductivity is sufficient for classification or monitoring purposes. Small surface coil RF probes are ideal for classifying large and irregular samples. For example, in food science, this technique could be used to differentiate soft- and hard-shell lobster based on their saline water content [38]. The pipe flow results could be employed to measure real-time conductivity in the electrolyte of redox flow batteries [2].

Because the method relies on the change in Q -factor of the RF resonator, it is inherently sensitive to conductivity, even in complex samples, provided that consistent positioning and probe loading are maintained. Conductivity measurements using RF probe loading has the potential to complement, and in some cases surpass, conventional conductivity measurement techniques, offering a reliable, non-invasive alternative for conductivity sensing in diverse applications.

Data availability statement

All data that support the findings of this study are included within the article (and any supplementary files).

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